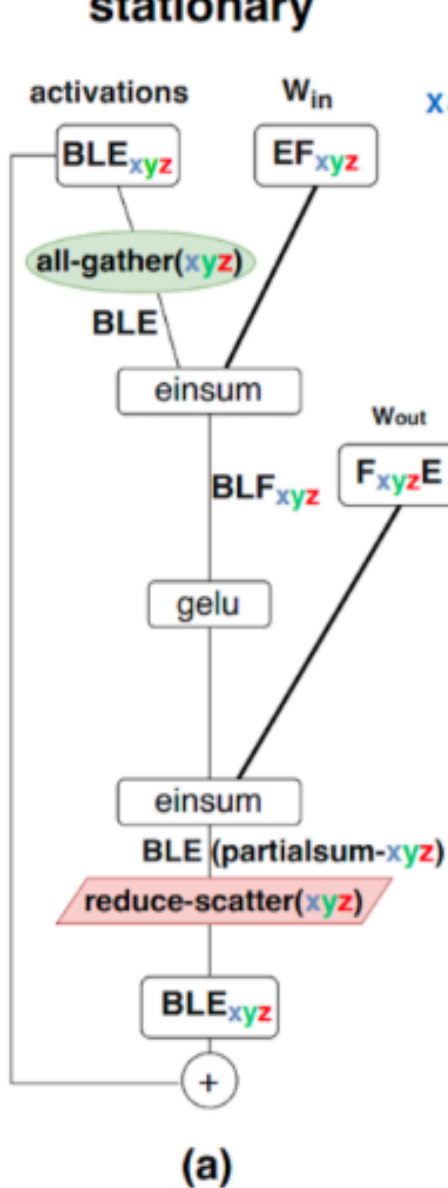


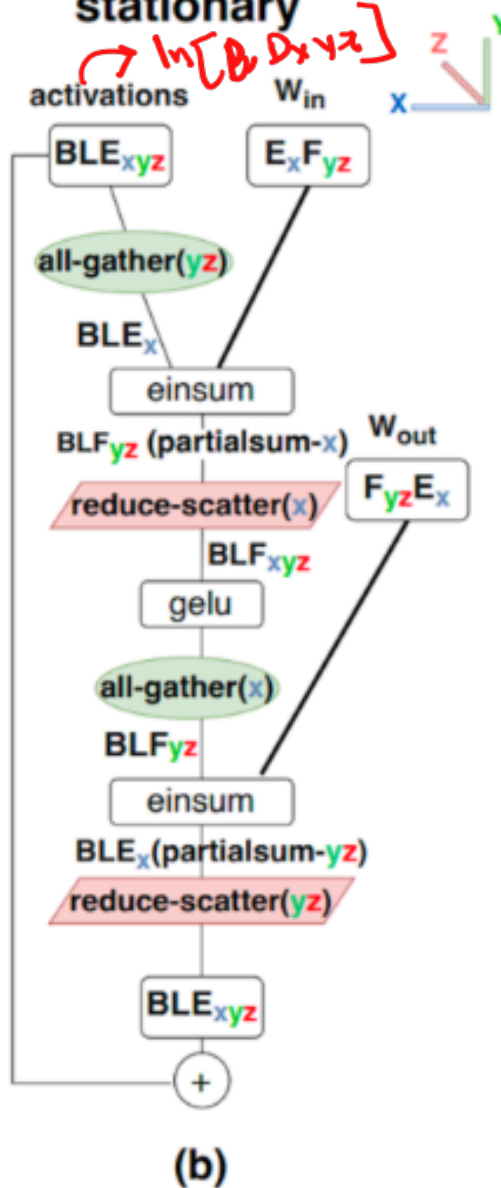
$W_{in} [0, F_N]$
 $W_{out} [F_N, 0]$

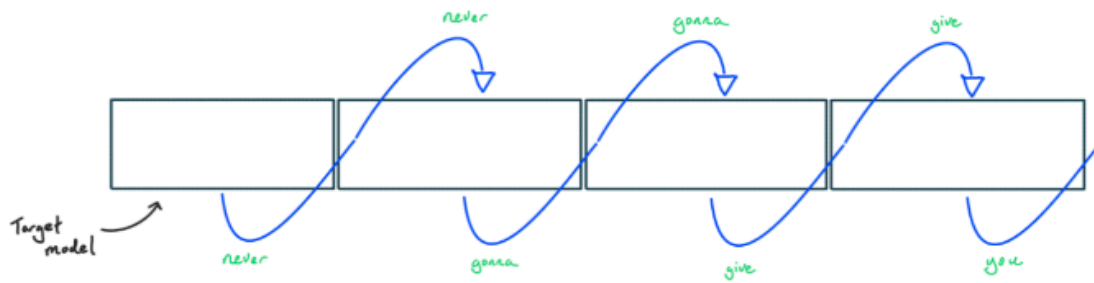
$W_{in} [D_x, F_{yz}]$
 $W_{out} [F_{yz}, D_x]$

1D weight-stationary

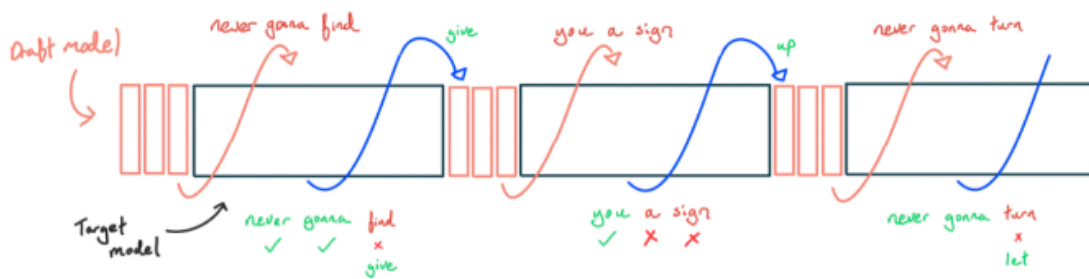


2D weight-stationary





With speculative sampling, we use a smaller, cheaper model to generate tokens and then check the result with the big model. This is easiest to understand with *greedy decoding*.

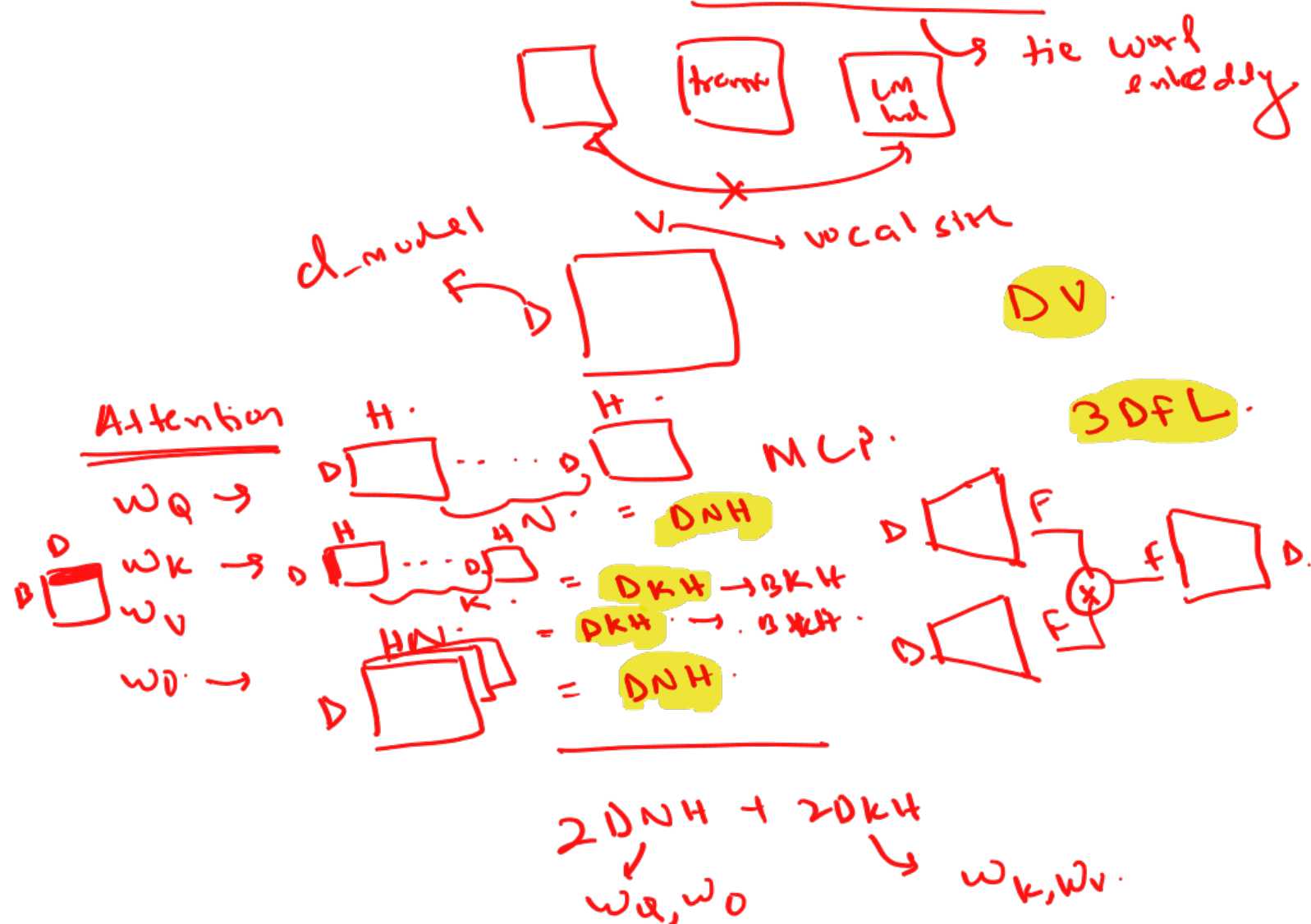


hyperparam	value
L (num_layers)	64
D (d_model)	4,096
F (ffw_dimension)	16,384
N (num_heads)	32
K (num_kv_heads)	8
H (qkv_dim)	256
V (num_embeddings)	32,128

GQA

32 → query heads
8 → kv heads.

Question 1: How many parameters does the above model have? How large are its KV caches per token in int8? You can assume we share the input and output projection matrices.



$$DV + \underline{3DFL} + 2BNHL + 20KHL$$

$$DV + DL(3F + 2NH + 2KH)$$

$$4096 \times \underline{32128} + 4096 \times 64 \left(\dots \right) \approx 13.468$$

kv
Cache

2

↓
K and V

= 2

num-hits
x K x H x L x 1
↓
sq. dist
leg. n
1 + 2

$$x 8 x 756 x 64 \approx 26263$$

Question 2: Say we want to serve this model on a TPUv5e 4x4 slice and can fully shard our KV cache over this topology. What's the largest batch size we can fit, assuming we use int8 for everything and want to support 128k sequences? What if we dropped the number of KV heads to 1?



$$\text{KV cache} = 262 \text{ kB} \times 128 \text{ k} \approx 33.5 \text{ GB}$$

$$P = 13.4 \text{ GB}$$

$$B \times \text{KV cache} + P = \text{total mem.}$$

$$B \times \text{KV cache} + P = 16 \times 16$$

$$\Rightarrow B = \frac{256 - 13.4}{33.548}$$

$$\approx 7.7$$

$$\approx 56$$

(# kv heads = 1)

Question 3: How long does it take to load all the parameters into the MXU from HBM assuming they're fully sharded on a TPU v5e 4x4 slice? Assume int8 parameters. *This is a good lower bound on the per-step latency.*

HBM



12.4 GB

$$\frac{12.4}{16 \times w_{hbm}} \approx \frac{18.4}{16 \times 3.2e''}$$

$$T_{\text{step}} = T_{\text{kv cache}} + T_{\text{hbm}}$$

$$= \frac{13 \cdot \text{kv cache}}{P} + \frac{P}{16 \cdot \text{whbm}}$$

$$= \frac{7 \cdot 33.5 \text{e}^9}{16 \cdot 3.2 \text{e}^{11}} + \frac{13 \cdot 4 \text{e}^9}{16 \cdot 3.2 \text{e}^{11}}$$

Question 4: Let's say we want to serve this model on a TPUv5e 4x4 slice using int8 FLOPs and parameters/activations. How would we shard it for both prefill and decode? *Hint: maybe answer these questions first:*

1. What does ICI look like on a 4x4?

2. What's the roofline bound on tensor parallelism?

3. How can we shard the KV caches?

$$\frac{N-1}{N} \times \frac{V}{W_{\text{ICI}}} = \frac{3}{4} \times \frac{V}{W_{\text{ICI}}}$$

$\frac{3}{4}(4)$

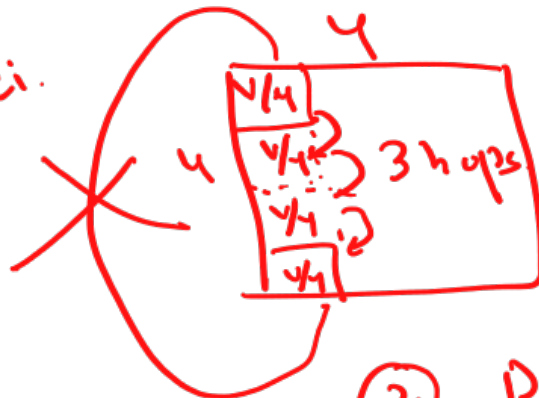


For this sharding, what is the rough per-step latency for generation?

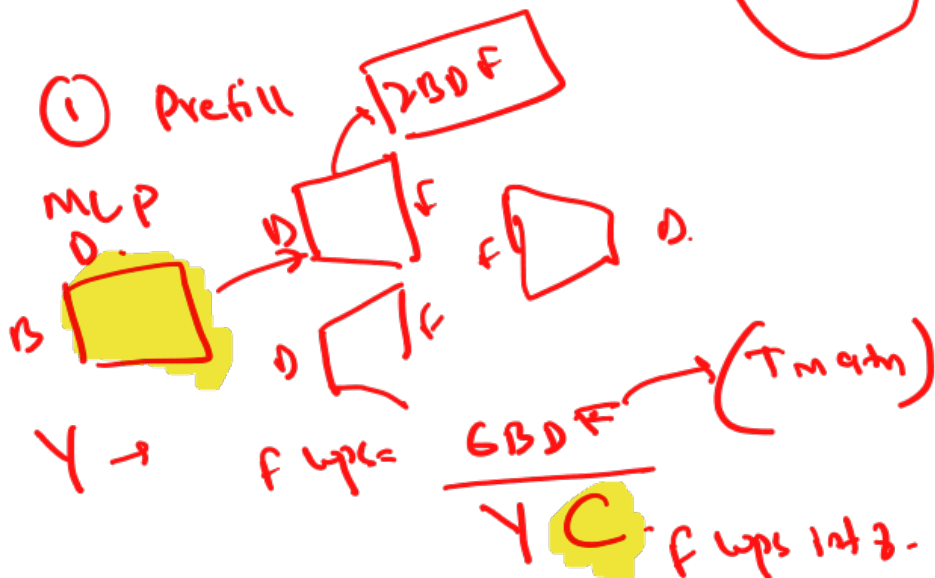
$N=4$

TPU v5e (no wraparound)

$$\frac{N-1}{N} \times \frac{V}{W_{\text{ICI}}} \times \frac{1}{W_{\text{ICI}}}$$

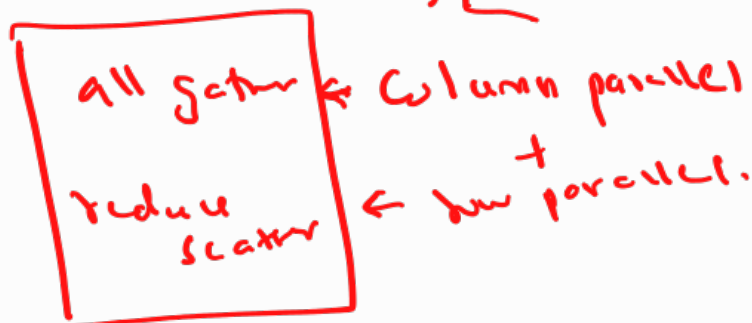
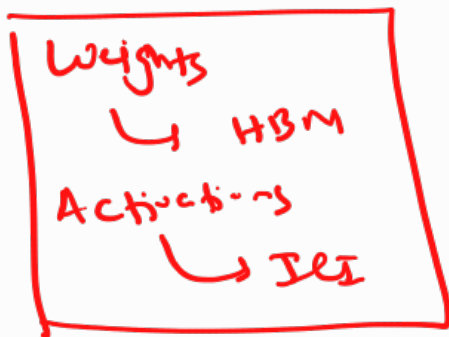


(2) Decode



TP → more activations.

(2 communications
all gather + reduce scatter)



$\frac{3}{4} \frac{V}{w_{ci}}$

$V = BD$

$\xrightarrow{\text{all gather} + \text{reduce scatter}} = 2 \times \frac{3}{4} \frac{BD}{w_{ci}} \approx \frac{3}{2} \frac{BD}{w_{ci}} \rightarrow \text{Turns}$

$\frac{2}{6} \frac{BD}{F} \geq \frac{3}{2} \frac{BD}{w_{ci}}$

$\leftarrow \gamma \approx 8 \rightarrow \text{tensor parallel}$

Prefill $\rightarrow$ 8 way to new parallel (2 way) 8 way shard.

Sequence parallel

Dewide (memory bound)

$T_{ci} = \frac{3}{2} \frac{BD}{w_{ci}}$

$\frac{P}{w_{hbm}} \xrightarrow{3 \text{ MLPs}} \frac{3BF}{w_{hbm}} \rightarrow 3 \text{ matrices.}$

$T_{hbm} \geq T_{ci}$

$v_{hbm} \gg w_{ci}$

weights
 HBM $m \times v$

$\frac{3BF}{w_{hbm}} \geq \frac{3}{2} \frac{BD}{w_{ci}}$

$\gamma < \frac{1800}{B}$ (hide activations, transfer inside weight transfer)

$\gamma \approx \frac{1800}{B}$

$$Y < \frac{1100}{16} \quad B \approx 7$$

$$\boxed{Y < 156} \rightarrow 16 \text{ Shards (TPU)}$$

Question 5: Let's pretend the above model is actually an MoE. An MoE model is effectively a dense model with E copies of the FFW block. Each token passes through k of the FFW blocks and these k are averaged to produce the output. Let's use $E=16$ and $k=2$ with the above settings.

1. How many total and activated parameters does it have? *Activated means used by any given token.*
2. What batch size is needed to become FLOPs bound on TPU v5e?
3. How large are its KV caches per token? $\rightarrow$ does not change.
4. How many FLOPs are involved in a forward pass with T tokens?

Find 2 Tractional parameters

$$\text{total} = 3E DFL + 2DNH + 2DKH + DV.$$

$\rightarrow$ Hexpm. $\rightarrow$ weight $\rightarrow$ w_u, w_v

$= 212 \text{ GB.}$

$$\text{Activated} = \underline{\underline{3k DFL.}}$$

(2)

$$T_{\text{matm}} = \frac{2Bk DF}{C}$$

$\rightarrow$ batch size $\rightarrow$ weight $\rightarrow$ # actual ops

$$T_{\text{wmm}} = \frac{2EDF}{w_{\text{mm.}}} \quad \left. \vphantom{\frac{2EDF}{w_{\text{mm.}}}} \right\} \text{weight body}$$

$$T_{\text{matm}} \geq T_{\text{wmm}} \quad \rightarrow \text{+ pm vce}$$

$$\frac{Bk}{E} \geq \frac{C}{w_{\text{mm.}}}$$

$$\geq 240$$

$\rightarrow$ batch size around 240

$$\boxed{B \geq 3 \times 240}$$

Question 6: With MoEs, we can do "expert sharding", where we split our experts across one axis of our mesh. In our standard notation, our first FFW weight has shape $[E, D, F]$ and we shard it as $[E_Z, D_X, F_Y]$ where X is only used during training as our FSDP dimension. Let's say we want to do inference on a TPU v5e:

↳ no weight sharding

1. What's the HBM weight loading time for the above model on a TPU v5e 8×16 slice with $Y=8$, $Z=16$? How much free HBM is available per TPU?
2. What is the smallest slice we could fit our model on?

$$16 - \frac{212}{8 \times 16}$$

$$\approx 1.66 \text{ GB}$$

$$(16 - 1.66) \text{ free}$$

16 GB

□ HBM.

T_{weight}

$$\frac{1.66}{3.2 \times 10^{11}} \approx 2 \text{ ms}$$

$$\text{Q2} \rightarrow \frac{212e^9}{16e^9 N} \geq 1$$

$$N \approx 14$$

$$256 - 212 \approx (44 \text{ GB})$$

↓
(KV cache)

(4×4)
or
(16 chips)

↓
256 GB
many

Question 7 [2D model sharding]: Here we'll work through the math of what the [ESTI paper](#) calls 2D weight-stationary sharding. We describe this briefly in Appendix B, but try doing this problem first to see if you can work out the math. The basic idea of 2D weight stationary sharding is to shard our weights along both the D and F axes so that each chunk is roughly square. This reduces the comms load and allows us to scale slightly farther.

Here's the algorithm for 2D weight stationary:

2D weight sharding (64x64)

1. $\text{In}[B, D_X] = \text{AllGather}_{YZ}(\text{In}[B, D_{XYZ}])$
2. $\text{Tmp}[B, F_{YZ}] \{U_X\} = \text{In}[B, D_X] *_{D'} W_{\text{in}}[D_X, F_{YZ}]$
3. $\text{Tmp}[B, F_{YZ}] = \text{AllReduce}_X(\text{Tmp}[B, F_{YZ}] \{U_X\})$
4. $\text{Out}[B, D_X] \{U_{YZ}\} = \text{Tmp}[B, F_{YZ}] *_{F'} W_{\text{out}}[F_{YZ}, D_X]$
5. $\text{Out}[B, D_{XYZ}] = \text{ReduceScatter}_{YZ}(\text{Out}[B, D_X] \{U_{YZ}\})$

1D weight sharding

$$= \frac{2BD}{3wici} + \frac{2BD}{3wici} \cdot 3wici$$

all gather, reduce, scatter

$$T_{\text{Comms}} = 2B \frac{D}{X} \times \frac{1}{Yz} \cdot 2wici + 2 \times 2B \frac{F}{Yz} \times \frac{1}{wici} + 2BD \frac{F}{X} \times \frac{1}{2wici}$$

all reduce = 2x all gather

$$T_{\text{Computation}} = \frac{2B}{N} \frac{D}{X} \frac{F}{Yz} \times \frac{1}{C} = \frac{2BDF}{N^2 C} + \frac{2BDF}{N^2 C} = \frac{4BDF}{N^2 C}$$

$$T_{\text{wnms}} = \frac{2B}{w_{\text{ici}}} \left(\frac{D}{X} + \frac{2F}{\sqrt{2}} \right)$$

all reduce
+
all gather
+
reduce
scatter

$\sqrt{2} = N \cdot \frac{N}{X}$

$$\frac{\partial T}{\partial X} = -\frac{D}{X^2} + \frac{2F}{N} = 0$$

$$X = \sqrt{\frac{DN}{2F}} = \sqrt{\frac{N}{\frac{F}{2D}}}$$

$$\sqrt{2} = \sqrt{2N}$$

$(X) \sqrt{2} = N$

$\downarrow$
 $\sqrt{N/2}$

$T_{1D \text{ comms}}$

$$\frac{4BD}{3w_{\text{ici}}}$$

$T_{2D \text{ comms}}$

$$\frac{\sqrt{128} BD}{\sqrt{N} w_{\text{ici}}}$$

$$N > 128 \cdot (3/4)^2 = 72$$

($N > 72$)
(2D will take less time)

CHAPTER 8

hyperparam	value
n_{layers} (L)	80
d_{model} (D)	8,192
d_{ff} (F)	28,672
n_{heads} (N)	64
$n_{\text{kv heads}}$ (K)	8
d_{qkv} (H)	128
$n_{\text{embeddings}}$ (V)	128,256

Question: How large are LLaMA 3-70B's KV caches per token? You can assume we store them in int8. This determines how large our batch size can be on a given topology.

$$2 \times L \times \underbrace{K}_{\substack{\text{K and V} \\ \text{# of kv}}} \times H = 2 \times 280 \times 8 \times 128 \approx \underline{\underline{1000}}$$

$$70 \times 2 = 70e^9 \text{ or } 70 \text{ GB.}$$

→ 70 GB.

Question: Let's say we want to serve L3 70B at batch size 32 and 8192 sequence length with everything (params and KVs) in int8. How much total memory will this use? What's the smallest slice we could serve this on?

← 13. KV cache + P.

(HBM = 16 GB)
per chip.

$$32 + 8192 \times 16013 + 70$$

$$= 41.9 \text{ GB} + 70 = 112 \text{ GB.}$$

$$\frac{112}{16} \approx 7 \text{ TPUS}$$

$$(4 \times 2)$$

$$(4 \times 4)$$

Tensor parallelism.

but each active trans.

Question: At this batch size and quantization on a TPU v5e 4x2 roughly what latency would we expect per decode step? What throughput (tokens / sec / chip). What about a 4x4? Assume we perform our FLOPs in bfloat16 and everything is fully sharded.

After memory load

MCP.

$T_{step} =$
 $C_{kv} = \text{cache size}$

$$\frac{B \cdot C_{kv}}{n \cdot W_{kv}}$$

$$+ \max \left(\frac{2B P_{mat}}{n \cdot C}, \frac{P_{mat}}{n \cdot W_{kv}} \right)$$

count size

Refill compute word
 Decode + memory word

$$\frac{B \cdot C_{kv}}{n \cdot W_{kv}} + \frac{P_{mat \text{ size}}}{n \cdot W_{kv}}$$

$$T_{step} = \frac{112 \text{ GB}}{7 \times 7.2 \text{ e}''} \approx 17 \text{ ns}$$

$$\text{throughput} = \frac{B}{17 \text{ ns}} \approx 1382 \text{ tokens/sec}$$

$$\text{per chip} = \frac{1382}{8} \approx 172.75 \text{ tokens/sec/chip}$$

(Win, want)

$Y \rightarrow$ tensor parallelism
channel
dimension

$$T_{matn} = \frac{2BDF}{YC} + \frac{2BDF}{YC} = \frac{4BDF}{YC}$$

$T_{wms} =$
(1CS)

$4BD$

2x wici -
bidirectionally

$T_{matn} > T_{wms}$

all gather

$\rightarrow$ rank scatter

tpu v1e $\rightarrow$ X
bidirectional

$$Y \leq F / 2000$$

$$Y \leq \frac{2F}{2000}$$

$$Y \leq 13$$

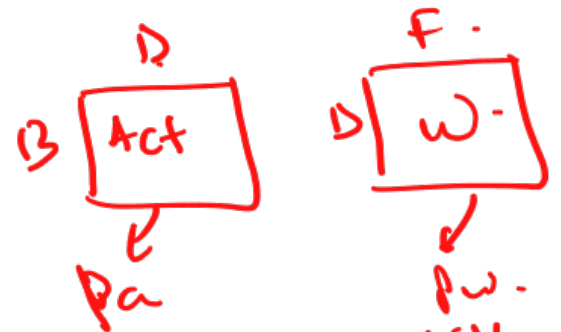
$\{X, Y\}$

Thinking about throughput

Let's spend a little time thinking purely about throughput. When we optimize for throughput, we want to be compute bound, meaning we come close to utilizing all the TPU MXU capacity. Typically that means we want the batch size to be as large as possible, so we are doing as much work as possible.

Question: On TPU v5e, using bfloat16 weights and activations, how large do our batch sizes need to be for us to be compute-bound in our matmuls? What if we do int8 weights but perform our FLOPs in bfloat16? What about int8 weights with int8 FLOPs?

$$C_{\text{int8}} = 2C_{\text{bfloat16}}$$



$$T_{\text{matm}} = \frac{2BDF}{C}$$

$$T_{\text{Wmatm}} = \frac{\frac{P_a}{8}BD + \frac{P_w}{8}DF + \frac{P_a}{8}DF}{1}$$

When

$$B \ll C < D, F$$

$$\frac{P_w DF}{8} \xrightarrow{\text{bfloat16}} \frac{2DF}{C} \text{ when } \frac{DF}{C} \text{ when } \text{int8}$$

$$\frac{2BDF}{C} > \frac{2DF}{C} \text{ when } B > 1$$

$$\frac{2BDF}{C} > \frac{DF}{C} \text{ when } B > 1$$

$$C_{\text{int8}} = 2C_{\text{bfloat16}}$$

Question: What is the smallest TPU v5e topology we could serve LLaMA 3-70B on using bfloat16, int8, and int4 (both KVs and parameters) with 8k context? *You can think of KV caches as negligibly small for this one.*

$$B \cdot C_{kv} + P.$$

→ kv cache

$$\rightarrow 3192 \times 26 \text{ KHz} \times 70 \times P$$

$$3192 \times 2 \times 20 \times 3 \times 128 \\ (1280 \times 128 \times 8192)$$

$$P = 2 \text{ (bf16)}$$

$$P = 2 \text{ (int8)}$$

$$P = \frac{1}{2} \text{ (int4)}$$

Question: Assume we use the largest batch size that fits on these topologies, what latency could we expect for each generate step?

$$\begin{aligned}
 & \frac{(B \cdot C_{KV} + P)}{n \cdot W_{HLM}} \rightarrow \frac{n \cdot H_{LM}}{n \cdot W_{HLM}} \\
 & = \frac{n \cdot 16}{n \cdot W_{HLM}} \\
 & = \frac{16}{8.2e''} \\
 & \sim 1.9 \text{ ms.} \\
 & \downarrow \\
 & \text{one token}
 \end{aligned}$$

Question: For each of these, what throughput per chip does this give us (in terms of queries / chip)? You can assume our median decode length is 512 tokens.

$$\left(\overset{\text{kv cache}}{B \cdot C_{kv}} + P_{\text{size}} \right) \leq N \cdot 16$$

$$B \cdot C_{kv} \leq N \cdot 16 - P_{\text{size}} P_w \quad \begin{matrix} \nearrow 2 \rightarrow 16 \cdot 16 \\ \nearrow 1 \rightarrow 16 \cdot 16 \end{matrix}$$

$$C_{kv} = 2 \times \text{LKH} P_{\text{cuv}}$$

$$= 2 \times 8192 \times \text{LKH} P_{\text{cuv}} \approx 1.3498 \cdot P_{\text{cuv}}$$

$$B \leq \frac{N \cdot 16 - P_{\text{size}} P_w}{P_{\text{cuv}} 1.3498}$$

int 2.

$$B \approx 43$$

throughput = $\frac{B}{T_{\text{step}}} = \frac{B}{19 \text{ms}}$

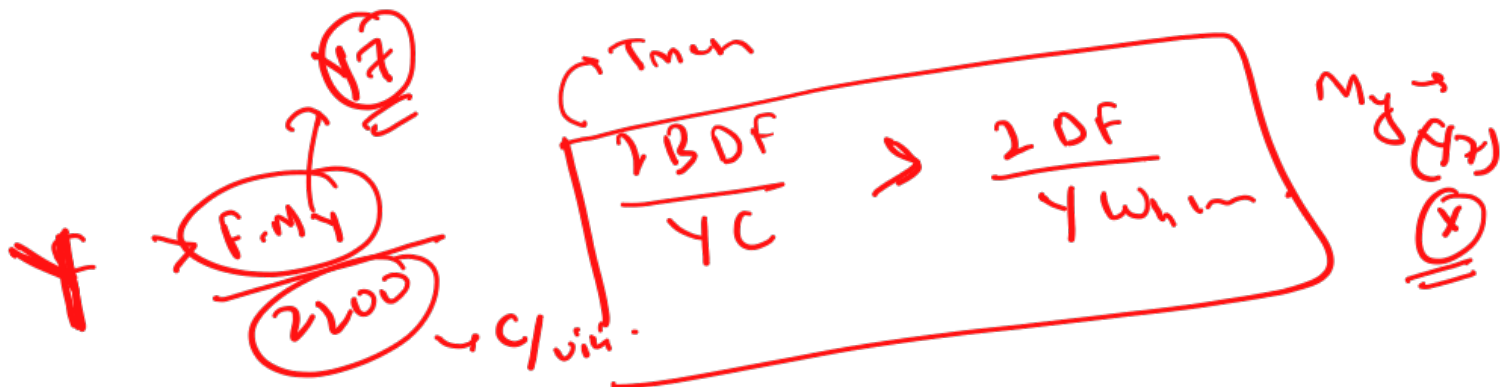
assuming 512 = $\frac{B}{512 \times 19} \times 1$

Question: How would our peak throughput change if we doubled our topology for each of the above examples?

$$B = \frac{16N - 70}{1.31e9} \quad \rightarrow \text{into m/s}$$

$$\text{throughput / chip} = \frac{\frac{16N - 70}{1.34e9}}{512 \times 19ms \times N} \propto \left(\frac{1}{N} \right)$$

$\propto \text{PS/chip.}$



Question: Now let's dig into the question of sharding. Let's say we wanted to serve in bfloat16 on a TPU v5e 4x8. What sharding would we use for our model on a TPU v5e 4x8 during generation? Can we avoid being communication bound?

Activations → TPU's ICI

weight → (HQM → MxV) → get weights

Flops → ⊗

$D_x \rightarrow 1$

B Act $\otimes$

F_y
 O W

$[B, D_x] \rightarrow [B, O]$

many iterations

T_weight

T_activation

all gather
bfloat16

T_weight

$= \frac{2 D F}{W_{h,m}}$

2 B D
W_{i,c}

or $2 D \frac{F}{Y} \times \frac{1}{W_{i,c}}$

(X) T_flops = $\frac{2 B D F}{Y C}$

$\frac{2 D F}{Y W_{h,m}} > \frac{2 B D}{W_{i,c}}$

$Y < \frac{F W_{i,c}}{W_{h,m} B}$

$$W_{min} \approx 3 \text{ vici}$$

$$Y \leq \frac{F}{8B}$$

Question: Assume we achieve a 40% FLOPs utilization during prefill. How long will a prefill of length 8192 take on 16 TPU v5e chips?

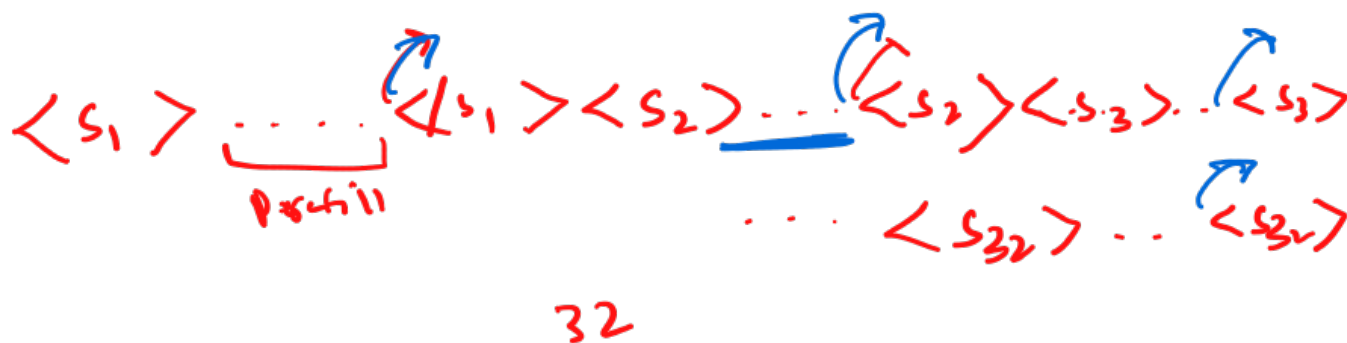
$$\frac{\text{FLOPs}}{\text{Acceleration time}} = \frac{2 \times 70e^9 \times B}{16 \times 1.97e^{14} \times \frac{40}{100}}$$

find 28ms/param
 int8

$$\approx 0.91s.$$

Question: Assume we have a median prefill length of 8192 tokens and a median decode length of 4096 tokens. Say we have a generate batch size of 32. On average how many sequences finish decoding per step? On average how many tokens are evicted from our KV cache each step?

Continuous batching



$$G = \underline{\underline{4}}$$

$$B = 2$$

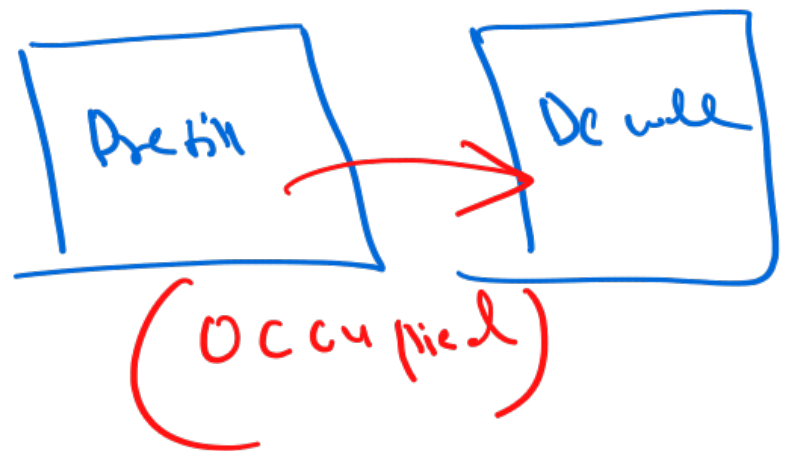
each slot finishes one sequence every 4 steps

$$B/G = 2/4 \text{ or } 0.5 \text{ per step}$$

$$\frac{B}{G}$$

$$\frac{32}{4096} \left(4096 + 8192 \right)$$

Question: Assume we do disaggregated serving with a median prefill length of 8192 and a median decode length of 512. Assume the prefill and generate latencies calculated above in bfloat16. What ratio of prefill:generate servers will you need to keep both fully saturated.

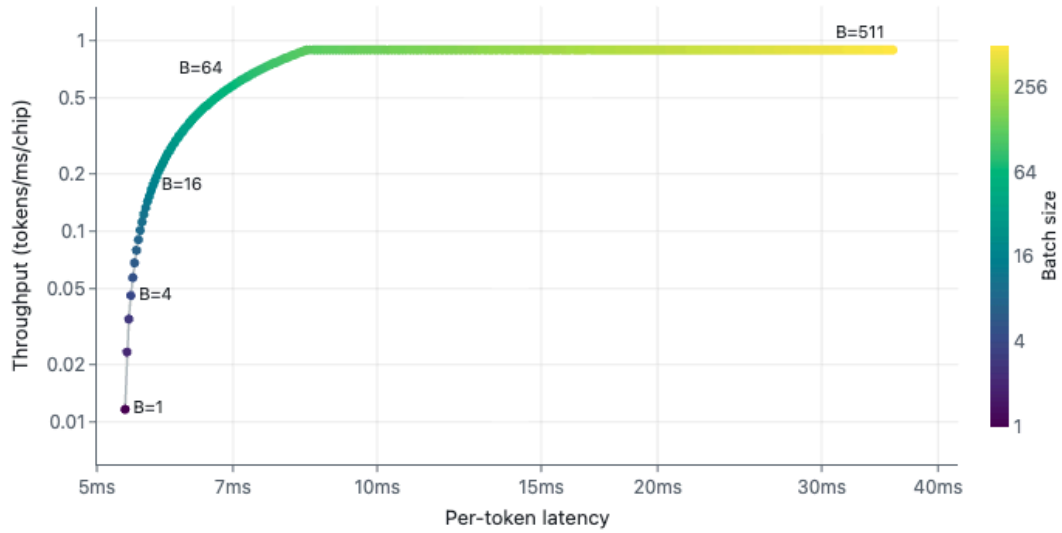


Prefill = one prompt per prefill req
 $\frac{1}{0.91} \approx 1.1 \text{ req/s}$

Generate = $\frac{32}{0.019 \cdot 512} \text{ req/s} \approx 3.3 \text{ req/s}$

$\frac{P}{0.91} = \frac{329}{0.019 \cdot 512} \Rightarrow P/G \approx 3$
 $P \approx 3G$

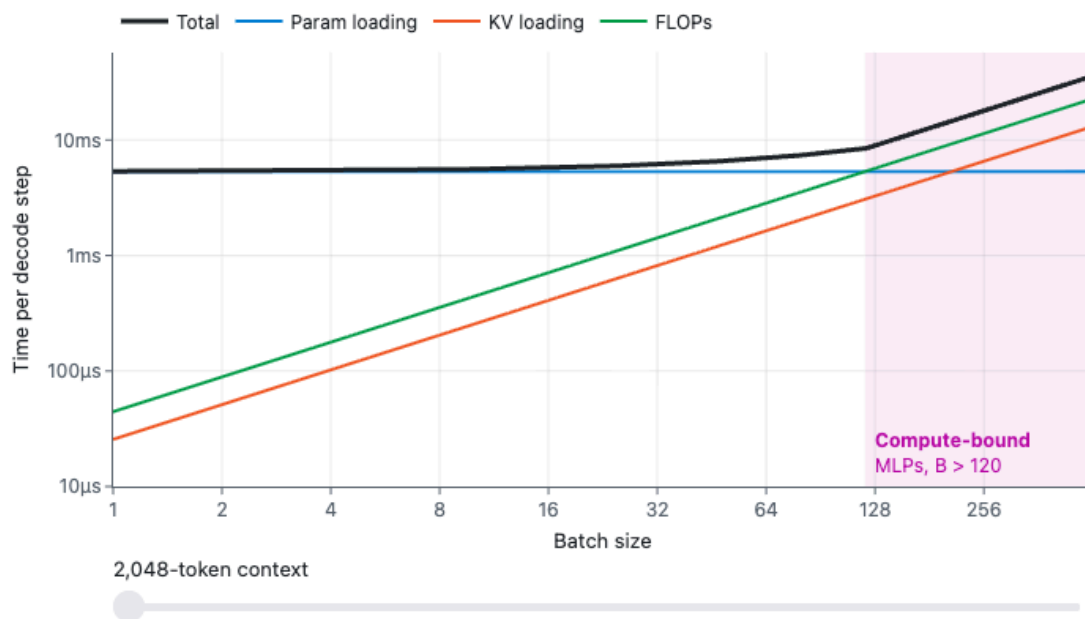
Latency vs. throughput for LLaMA 3-70B on TPU v5e 4x4



2,048-token context



Decode step time for LLaMA 3-70B on TPU v5e 4x4



Question 1: How many FLOPs does each forward pass for LLaMA 3-405B use per-token?

Assuming we're FLOPs bound, what is a lower bound on a single forward pass on N chips on TPU v5e? What if we're comms bound? *Ignore the fact that the model does not fit on a single chip.*

Question 2: Assume we want to serve LLaMA 3-8B with BS240 using int8 weights and int8 KV caches. How many bytes are used by (a) model parameters (b) KV caches and (c) peak working activations (roughly)? What's the smallest topology we can run this on?

Question 3: How would you serve LLaMA 3-405B on TPU v5e? Assume int8 weights and bfloat16 FLOPs. Let's say we have a firm limit of 15ms / token, what's the highest throughput configuration we could achieve? What is the theoretical minimum step time?